

Sardar Patel Institute of Technology
(Electronics & Telecommunication Department)



Mobile Wireless Communication

Experiment 2

Name: Rajas Mangirish Wagle

Class: TE(EXTC)

UID: 2023200129

Batch: B4

TITLE: Fixed Channel Assignment Technique

AIM:

Use Any Programming Language like C,C++,Python, Java etc,

1. Assume Total Number of Channels(50-100).
2. Assume 10-15% of total Channels are reserved for Control channels.
3. Remaining are Voice/Data channels.
4. Distribute Control and Voice Channels in 7,9,13 size of Cluster.
5. Display results of Control and Voice channels separately in matrix format.

THEORY:

A channel set is a collection of channels that could be assigned to one cell. In fixed channel assignment (FCA), each cell assigns its own frequency channel to the mobile subscribers within its cell. Channel assignment is primarily based on causing least cochannel and adjacent channel interference in the cellular system. The channel assignment for each voice call is determined by MTSO on a short-term basis. In a fixed channel assignment, the set-up and voice channels are usually assigned to the cell-site for relatively long periods. Channels in a channel set are usually 21 channels apart and must meet minimum frequency spacing requirements of a multi-channel transmitter combiner. Channels are usually numbered in order of increasing frequency. Regardless of the number of channels in a channel set, the highest channel set is frequency-adjacent to the lowest channel set. There are certain advantages of fixed channel assignment such as fixed parameters (power, frequency) for transceivers, and reasonably good performance under uniform and/or high traffic loads as cells independently decide their channel allocation decisions. However, the number of channels in a channel set has to be at least seven for hexagonal cellular configuration. At run-time, channel borrowing may be necessary but the number of channels in a cell dictates the upper limits for channel borrowing. This results in a major problem of hot spots or localised congestion with fixed channel assignment.

CHANNEL SHARING:

When a particular cell needs more number of channels in order to meet the increased traffic to be served, the channels of another sector at the same cell-site can be shared to meet the short-term overload traffic. In a sectoral cellular system configuration, channel sharing can be done from one of the two adjacent sectors of the neighbouring cells. Shared channels can be returned back as and when the channels become available in the shared sector. This scheme is known as an ordered channel-assignment scheme with rearrangement. Another alternate scheme is channel assignment with sharing and reassignment. This scheme ensures that channel-sharing arrangement causes minimum impact on call-blocking probability in neighbouring cells. Reassignment of shared channels is done to provide maximum assistance to the neighbouring cells to meet the temporary increased traffic demand. The channel can also be ordered based on which channels provide better performance. The configuration of

transmitter channel combiner at the cell-site should be flexible in real time so as to adapt to the increased number of transceivers for an increased number of channels at a time. Channel sharing always increases the trunking efficiency of the channels. The difficulty in implementing an effective channel-sharing scheme is that all cell locations may not have non parent cell-sites, and there may be low quality connections to cell-sites which are not parents. The channels may be shared between adjacent cells. It should be noted that channels are not re-allocated but re-used by a different cell. Channel sharing may be considered as a temporary channel-borrowing strategy between one cell with the cell site of a neighbouring cell. Generally, non-parent cell-sites are chosen based on close proximity. Moreover it requires local channel management between adjacent cells. Also, it requires precise control of transmitted power of mobile equipment and cell-sites because there is increased probability of adjacent channel interference. There are relative advantages and disadvantages of different channel-sharing schemes in terms of proper utilisation of total available channels, total carried traffic by the system, and assignment complexity. Usually, decisions are made based on the system parameters and traffic behaviour.

CODE:

```
import math
```

```
#taking user input for total number of channels:
```

```
channel = int(input("Enter the number of Total Channels: "))
```

```
#user input for cluster size
```

```
cluster = int(input("Enter the cluster size: "))
```

```
#after entries, take percentage of control channels to be reserved from number of channels:
```

```
while True:
```

```
    ctrl_channel_per = int(input("Enter percentage of control channel (10 - 15 %): "))
```

```
    if 10 <= ctrl_channel_per <= 15:
```

```
        break
```

```
    else:
```

```
        print("Invalid input, please enter a value between 10 and 15.")
```

```
ctrl_channel = math.floor((ctrl_channel_per / 100) * channel)
```

```
print(f"\nControl channels reserved: {ctrl_channel}")
```

```
#control channels definition based on the range of reserved channels.
```

```
def control_channel_allocation(cluster, ctrl_channel):
```

```
    columns = math.ceil(ctrl_channel / cluster)
```

```
# Initializing matrix with zeroes initially
```

```
matrix = [[0 for _ in range(columns)] for _ in range(cluster)]
```

```

# Initializing channel number as an index
channel_num = 1

for col in range(columns):
    for row in range(cluster):
        matrix[row][col] = channel_num
        channel_num += 1
        #Return to initial channel if channel number exceeds number of control channels
pre-allocated.
        if channel_num > ctrl_channel:
            channel_num = 1
    return matrix

# function for defining the traffic channels based on the remaining channels
def traffic_channel_allocation(cluster, channel, ctrl_channel):
    traffic_channels = channel - ctrl_channel
    columns = math.ceil(traffic_channels / cluster)
    matrix = [[0 for _ in range(columns)] for _ in range(cluster)]
    channel_num = ctrl_channel + 1

    for col in range(columns):
        for row in range(cluster):
            if channel_num <= channel:
                matrix[row][col] = channel_num
                channel_num += 1
    return matrix

allocation_matrix = control_channel_allocation(cluster, ctrl_channel)

#printing the control channels:
print("\nControl Channel Allocation Matrix:")
print(f"Rows (Cells): {cluster}")
print(f"Columns: {math.ceil(ctrl_channel / cluster)}")
print("\nMatrix:")
for row in allocation_matrix:
    print(" [" + " ".join(f"{num:2d}" for num in row) + "]")

traffic_matrix = traffic_channel_allocation(cluster, channel, ctrl_channel)

#printing the traffic channels:
print("\nTraffic Channel Allocation Matrix:")

```

```

print(f"Rows (Cells): {cluster}")
print(f"Traffic Channels: {channel - ctrl_channel}")
print(f"Columns: {math.ceil((channel - ctrl_channel) / cluster)}")
print("\nMatrix:")

```

```

for row in traffic_matrix:
    print(" [" + " ".join(f"{num:2d}" for num in row), "]")

```

OUTPUT(S):

1. For Cluster size of 7, Assuming a total of 100 channels, with 15% of those allocated for control channels(both user inputted),

```

Enter the number of Total Channels: 100
Enter the cluster size: 7
Enter percentage of control channel (10 - 15 %): 15

Control channels reserved: 15

Control Channel Allocation Matrix:
Rows (Cells): 7
Columns: 3

Matrix:
[ 1  8 15 ]
[ 2  9  1 ]
[ 3 10  2 ]
[ 4 11  3 ]
[ 5 12  4 ]
[ 6 13  5 ]
[ 7 14  6 ]

Traffic Channel Allocation Matrix:
Rows (Cells): 7
Traffic Channels: 85
Columns: 13

Matrix:
[ 16 23 30 37 44 51 58 65 72 79 86 93 100 ]
[ 17 24 31 38 45 52 59 66 73 80 87 94  0 ]
[ 18 25 32 39 46 53 60 67 74 81 88 95  0 ]
[ 19 26 33 40 47 54 61 68 75 82 89 96  0 ]
[ 20 27 34 41 48 55 62 69 76 83 90 97  0 ]
[ 21 28 35 42 49 56 63 70 77 84 91 98  0 ]
[ 22 29 36 43 50 57 64 71 78 85 92 99  0 ]

```

2. For Cluster size of 9, Assuming a total of 100 channels, with 15% of those allocated for control channels(both user inputted),

```
Enter the number of Total Channels: 100
Enter the cluster size: 9
➡ Enter percentage of control channel (10 - 15 %): 15
```

Control channels reserved: 15

Control Channel Allocation Matrix:

Rows (Cells): 9

Columns: 2

Matrix:

```
[ 1 10 ]
[ 2 11 ]
[ 3 12 ]
[ 4 13 ]
[ 5 14 ]
[ 6 15 ]
[ 7  1 ]
[ 8  2 ]
[ 9  3 ]
```

Traffic Channel Allocation Matrix:

Rows (Cells): 9

Traffic Channels: 85

Columns: 10

Matrix:

```
[ 16 25 34 43 52 61 70 79 88 97 ]
[ 17 26 35 44 53 62 71 80 89 98 ]
[ 18 27 36 45 54 63 72 81 90 99 ]
[ 19 28 37 46 55 64 73 82 91 100 ]
[ 20 29 38 47 56 65 74 83 92  0 ]
[ 21 30 39 48 57 66 75 84 93  0 ]
[ 22 31 40 49 58 67 76 85 94  0 ]
[ 23 32 41 50 59 68 77 86 95  0 ]
[ 24 33 42 51 60 69 78 87 96  0 ]
```

3. For Cluster size of 13, Assuming a total of 100 channels, with 15% of those allocated for control channels(both user inputted),

```
↔ Enter the number of Total Channels: 100
Enter the cluster size: 13
Enter percentage of control channel (10 - 15 %): 15

Control channels reserved: 15

Control Channel Allocation Matrix:
Rows (Cells): 13
Columns: 2

Matrix:
[ 1 14 ]
[ 2 15 ]
[ 3 1 ]
[ 4 2 ]
[ 5 3 ]
[ 6 4 ]
[ 7 5 ]
[ 8 6 ]
[ 9 7 ]
[ 10 8 ]
[ 11 9 ]
[ 12 10 ]
[ 13 11 ]

Traffic Channel Allocation Matrix:
Rows (Cells): 13
Traffic Channels: 85
Columns: 7

Matrix:
[ 16 29 42 55 68 81 94 ]
[ 17 30 43 56 69 82 95 ]
[ 18 31 44 57 70 83 96 ]
[ 19 32 45 58 71 84 97 ]
[ 20 33 46 59 72 85 98 ]
[ 21 34 47 60 73 86 99 ]
[ 22 35 48 61 74 87 100 ]
[ 23 36 49 62 75 88 0 ]
[ 24 37 50 63 76 89 0 ]
[ 25 38 51 64 77 90 0 ]
[ 26 39 52 65 78 91 0 ]
[ 27 40 53 66 79 92 0 ]
[ 28 41 54 67 80 93 0 ]
```

REFERENCE(s): Wireless Communication - Tata McGraw Hill(3rd Edition)

CONCLUSION:

I have successfully implemented a simulation for a Fixed Channel Allocation(FCA) setup in Python. I have read through the literature for Fixed Channel Allocation techniques, and hence have come to a conclusion that FCA is not the most suitable Channel allocation technique for a large user interface as it limits the concurrent number of users on a specific base station. Theoretically it is still relevant, however for the current magnitude of users reliant on 4G/5G Cellular communication, it is not suitable for usage.